Minimum Waiting Time

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The problem

- Minimum Waiting

Intuition

- Short queries should run first, because each query's duration is waited on by everyone after it.

Approach

- 1. Sort durations ascending
- 2. Each duration is waited on by the queries remaining after it
- 3. Add duration times queries left
- 4. Return the total

Java solution

```java
public class MinimumWaitingTime {
    public static int minimumWaitingTime(int[] queries) {
        Arrays.sort(queries);
        int total = 0;
        for (int i = 0; i < queries.length; i++) total += queries[i] * (queries.length - 1 - i);
        return total;
    }
}
```

Complexity

- Time: queries with durations run one at a time; choose an order minimising the total time all queries spend waiting. Shortest-first is optimal. Time: O(n log n)
- Space: O(1)

Recap

- Short queries should run first, because each query's duration is waited on by everyone after it.
- Complexity: queries with durations run one at a time; choose an order minimising the total time all queries spend waiting. Shortest-first is optimal. Time: O(n log n)